

LOFTUS, United Kingdom

WMO: 032750

Lat: 54.5628N

Lon: 0.8636W

Elev: 158

StdP: 99.44

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.1	-5.7	2.4	-0.9	-4.4	2.7	0.1	20.0	7.8	18.5	7.4	4.7	250	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	23.1	17.8	21.4	16.7	19.9	15.8	18.6	21.8	17.5	20.3	16.6	19.0	5.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.6	19.9	16.3	11.8	18.8	15.5	11.2	17.8	53.2	22.0	49.8	20.4	47.0	19.0	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	14.0	12.4	-4.2	26.7	1.6	1.9	-5.3	28.0	-6.2	29.1	-7.1	30.2	-8.3	31.5
			-5.0	21.1	1.4	1.7	-6.0	22.3	-6.8	23.3	-7.6	24.3	-8.7	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	4.0	4.3	5.8	7.4	10.2	12.7	15.0	15.2	13.3	10.1	6.5	4.5
	DBStd	4.83	2.54	2.82	2.85	2.88	2.76	2.59	2.48	2.41	2.37	2.61	2.67	2.91
	HDD10.0	916	187	160	134	87	32	4	0	0	1	31	108	172
	HDD18.3	3381	445	392	389	328	253	169	108	103	153	256	354	430
	CDD10.0	589	0	1	3	10	37	87	155	160	99	33	4	1
	CDD18.3	11	0	0	0	0	0	1	4	4	2	0	0	0
	CDH23.3	44	0	0	0	0	1	7	18	14	4	1	0	0
	CDH26.7	3	0	0	0	0	0	1	2	1	0	0	0	0
Wind	WSAvg	5.9	7.3	7.0	6.4	5.5	5.1	4.8	4.5	5.0	5.5	6.4	7.0	7.0
Precipitation	PrecAvg	705	61	50	40	51	47	64	57	68	57	65	78	67
	PrecMax	945	123	83	109	139	104	196	107	145	118	125	150	114
	PrecMin	503	13	6	2	3	11	9	16	9	16	18	25	23
	PrecStd	110	30	22	27	41	24	41	25	33	31	25	30	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.3	15.5	18.8	21.4	24.3	25.4	25.1	24.0	18.6	14.6	12.3
		MCWB	10.5	10.4	11.1	13.1	16.2	18.6	18.2	19.0	17.8	15.6	13.2	11.1
	2%	DB	10.3	11.1	12.9	15.6	18.3	21.0	23.0	22.5	20.6	16.5	12.9	11.2
		MCWB	9.0	9.2	9.9	11.6	14.5	16.6	17.6	17.7	16.2	14.4	11.8	10.1
	5%	DB	9.0	9.7	11.4	13.7	16.6	19.0	21.2	20.8	18.7	15.1	11.7	10.0
		MCWB	7.8	8.1	8.9	10.4	12.9	15.4	16.5	16.5	15.0	13.6	10.6	9.0
	10%	DB	7.9	8.6	10.1	12.2	15.0	17.4	19.6	19.3	17.2	13.9	10.5	8.8
		MCWB	6.8	7.0	8.0	9.3	12.0	14.3	15.7	15.6	14.1	12.5	9.6	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.7	11.7	13.7	17.1	20.1	19.6	20.5	18.6	16.2	13.5	11.5
		MCDB	11.6	12.5	14.0	16.7	20.1	23.0	23.8	22.4	22.2	17.9	14.2	12.1
	2%	WB	9.2	9.4	10.5	12.1	15.0	17.4	18.3	18.5	16.9	14.8	12.0	10.3
		MCDB	10.2	10.9	12.5	14.7	17.9	20.1	21.7	21.5	19.6	16.0	12.7	11.0
	5%	WB	8.0	8.2	9.2	10.7	13.5	15.8	17.2	17.4	15.7	13.8	10.8	9.1
		MCDB	8.8	9.5	11.0	13.3	15.9	18.3	20.1	19.9	17.9	14.9	11.5	10.0
	10%	WB	6.9	7.2	8.1	9.6	12.2	14.7	16.2	16.4	14.6	12.6	9.7	7.9
		MCDB	7.7	8.4	9.9	11.8	14.5	17.0	18.8	18.4	16.5	13.7	10.4	8.7
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.7	5.8	6.3	6.5	6.8	7.1	7.1	6.4	5.2	4.5	4.2
		MCDBR	5.5	5.9	7.5	9.0	9.3	9.6	9.8	9.1	8.9	6.4	5.5	5.4
		MCWBR	5.3	5.0	5.4	5.9	6.3	6.4	5.5	5.4	5.4	5.0	5.1	5.2
	5% WB	MCDBR	5.5	5.8	7.1	8.4	8.7	9.2	8.8	8.2	7.9	6.3	5.3	5.4
		MCWBR	5.4	5.1	5.6	5.9	6.3	6.7	5.6	5.7	5.4	5.3	5.1	5.3
Clear-Sky Solar Irradiance	taub	0.314	0.331	0.360	0.385	0.381	0.386	0.387	0.378	0.358	0.341	0.316	0.299	
	taud	2.346	2.411	2.342	2.295	2.358	2.365	2.366	2.406	2.462	2.473	2.402	2.313	
	Ebn at Noon	631	758	812	839	864	862	852	839	810	737	627	561	
	Edh at Noon	55	73	96	115	114	114	112	102	85	68	52	46	
All-Sky Solar Radiation	RadAvg	0.59	1.22	2.32	3.69	4.74	4.80	4.62	3.82	2.75	1.52	0.74	0.45	
	RadStd	0.08	0.14	0.30	0.39	0.26	0.41	0.39	0.25	0.25	0.13	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page